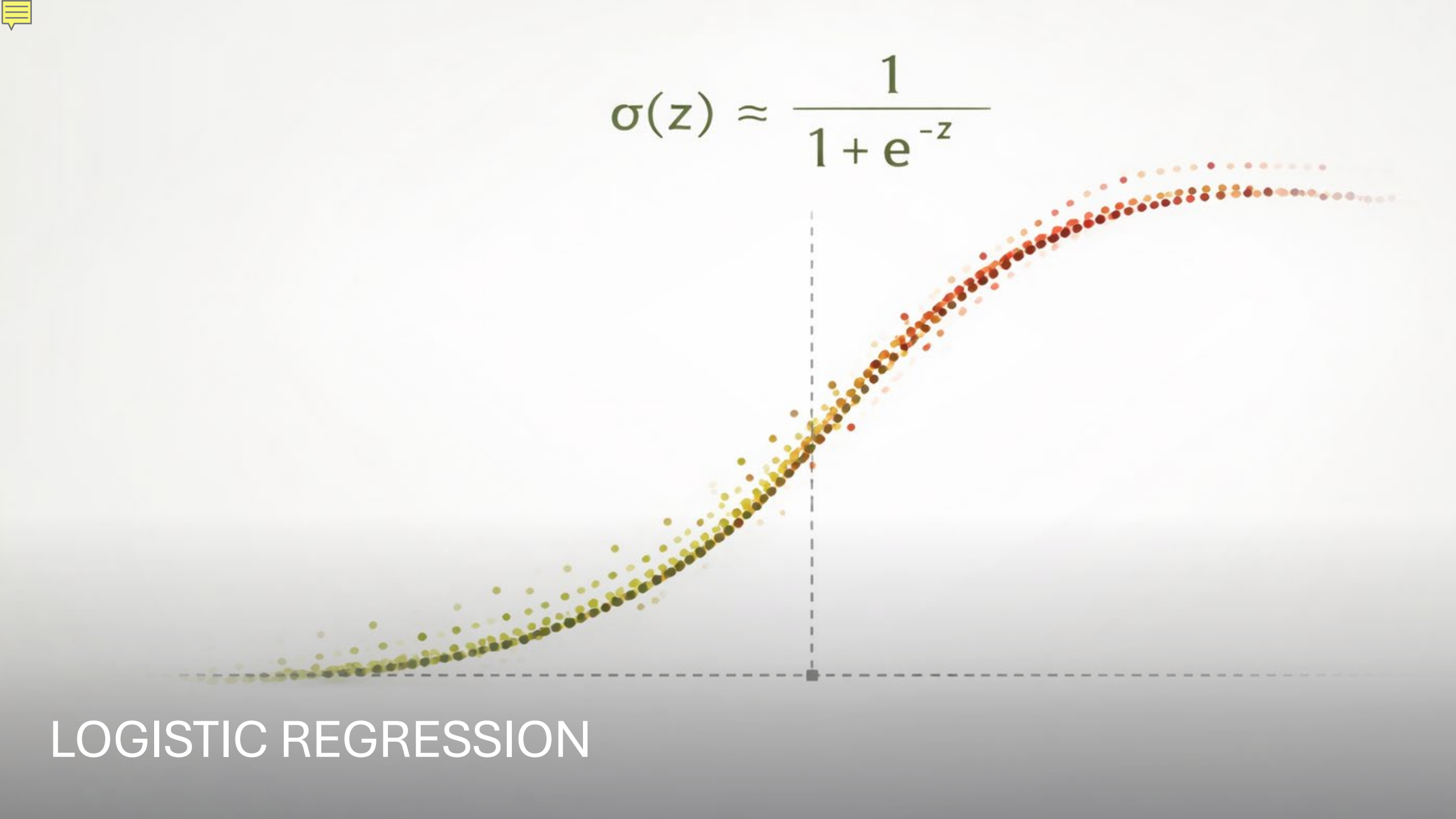
$$\sigma(z) \approx \frac{1}{1+e^{-z}}$$



LOGISTIC REGRESSION

PART 1 — SETTING UP THE MODEL

From a binary outcome to a probability model.

- Why is linear regression the wrong tool here?

  Unbounded predictions: $\hat{y} \in (-\infty, \infty)$, not $[0,1]$

  Wrong error model: assumes Gaussian noise, not Bernoulli

- We need a function that maps a real-valued score into $(0,1)$?

MOTIVATION: BINARY OUTCOMES

Goal: predict a binary outcome $y \in \{0,1\}$ from a feature vector $x \in \mathbb{R}^p$

Running example: predicting credit-card default ($y = 1$) from account balance.

Notation used throughout

- $x = (x_1, \ldots, x_p)$: features (predictors) measured on one observation
- $w \in \mathbb{R}^p$: weights (coefficients), one per feature
- $b \in \mathbb{R}$: bias (intercept)
- $p = P(y = 1 \mid x)$: probability the outcome is 1, given the features

FROM LINEAR SCORES TO PROBABILITIES

Linear predictor (score):

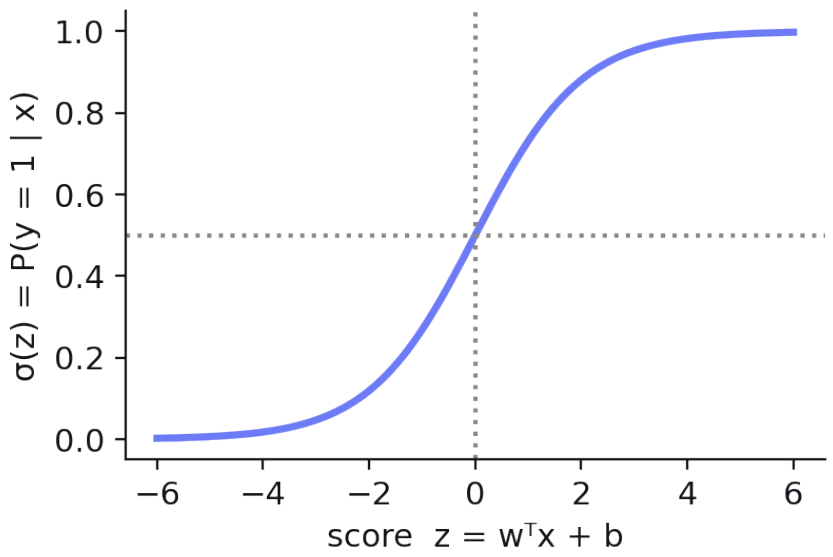
- $z = w^T x + b$
  - In linear regression, z is out output! (prediction)
  - Here, it is a **latent score**

We need a smooth, monotonic map from $\mathbb{R}$ into (0,1):

- $\sigma(z) = 1 / (1 + e^{-z})$
- Sigmoid or Logistic function

Model:

- $P(y = 1 \mid x) = \sigma(w^T x + b)$
- What is P(y = 0 | x)?
- By symmetry: P(y = 0 | x) = 1 − σ(z) = σ(−z)



LINEAR VS. LOGISTIC, SIDE BY SIDE

Both panels fit default probability against account balance.

Left: the linear fit leaves the [0,1] range, predicting negative probabilities at low balances. Right: the sigmoid fit stays in (0,1) and bends where default risk actually changes.

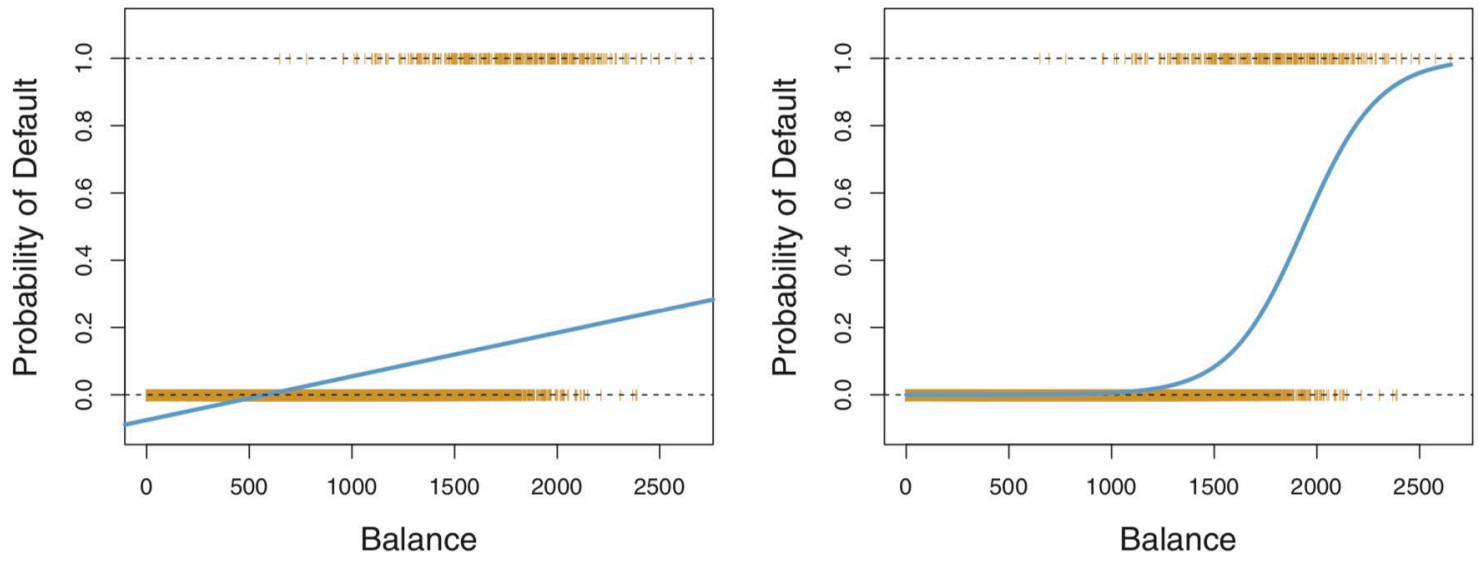


Figure from Introduction to Statistical Learning by James et al.

ODDS AND THE LOGIT

Odds

The ratio of success to failure probability:

$$\text{odds}(x) = p \,/\, (1 - p)$$

$$p \in (0,1) \;\Rightarrow\; \text{odds} \in (0, \infty)$$

A probability of 0.2 is odds of 0.25 — one default for every four non-defaults.

Odds rescale a probability onto (0, ∞), but they are still not linear in the features.

Log-odds (the logit)

The sigmoid and the logit are exact inverses — each undoes the other:

$$p = \sigma(w^T x + b) \;\Leftrightarrow\; \text{logit}(p) = w^T x + b$$

$$\text{where}\;\; \text{logit}(p) = \log(\, p \,/\, (1 - p)\,)$$

$$\text{and}\;\; \sigma(z) = 1 \,/\, (1 + e^{-z})$$

σ carries a log-odds score into a probability; the logit carries a probability back to the score.

Taking the log is what makes the scale linear — logistic regression is a linear model on the log-odds scale.

READING THE COEFFICIENTS

The log-odds is an ordinary linear equation:

$$\text{logit}(p) = \log(\, p \,/\, (1 - p)\,) = w^{T}x + b$$

So read w_j exactly as a coefficient in linear regression — the change in the response for a one-unit increase in x_j, holding the others fixed. The only difference is that the response is the log-odds.

Exponentiate to move to the odds scale:

$$\text{odds} = p \,/\, (1 - p) = \exp(w^{T}x + b)$$

A one-unit increase in x_j therefore multiplies the odds by $\exp(w_j)$ — the odds ratio.

Rules of thumb

- $w_j > 0$ raises the probability of $y = 1$; $w_j < 0$ lowers it; $w_j = 0$ means no effect
- $w_j = 0.7 \Rightarrow \exp(0.7) \approx 2$, so each unit of x_j roughly doubles the odds
- Default example: $w = 2.0$ for balance (in \$1,000s) $\Rightarrow \exp(2.0) \approx 7.4$, so each extra \$1,000 multiplies the odds of default by about 7.4

Note

Coefficients are not changes in probability — the same odds ratio moves a probability a lot near 0.5 and very little near 0 or 1.

PART 2 — FITTING THE MODEL

From a model to fitted values of w and b.

- What probability model generated the data?
- What objective does that model imply?
- How do we minimize it when there is no closed form?

THE BERNOULLI MODEL

Explicitly model the data-generating process:

$$y \mid x \sim \text{Bernoulli}(p), \quad p = \sigma(w^T x + b)$$

A Bernoulli random variable equals 1 with probability p and 0 with probability 1 − p — a single weighted coin flip. Each customer either defaults or does not; the model sets the bias of that coin.

Why this choice

- It matches the binary nature of the data
- It gives a principled, likelihood-based

FROM LIKELIHOOD TO CROSS-ENTROPY

- Likelihood: how probable the observed data are under the model. For one observation:

$$P(y \mid x) = p^{y} (1 - p)^{1-y}$$

  - The exponents just select the right factor: p when y = 1, and 1 − p when y = 0.

- **Log-likelihood for n independent observations:**

$$\ell(w, b) = \Sigma_i [\, y_i \log p_i + (1 - y_i) \log(1 - p_i) \,]$$

- **Loss = negative log-likelihood, also called cross-entropy loss:**

$$\mathcal{L}(w, b) = -\Sigma_i [\, y_i \log p_i + (1 - y_i) \log(1 - p_i) \,]$$

- Maximizing the likelihood is exactly minimizing cross-entropy. A customer who defaulted but was given $p \approx 0$ contributes a very large loss.

OPTIMIZATION I: THE PROBLEM

Objective:

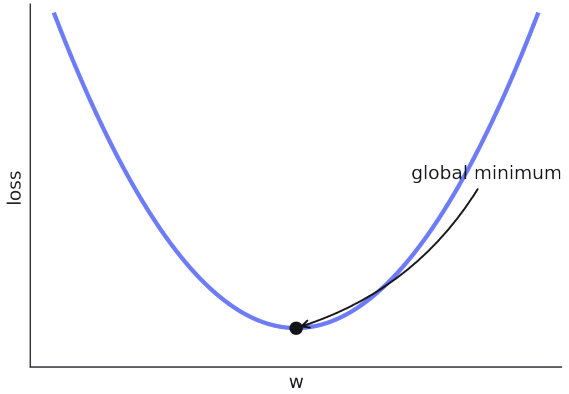
$$\min \text{ over } (w, b): \ \mathcal{L}(w, b)$$

Properties

- Convex — the loss surface is one bowl, so any local minimum is the global one
- No closed form — there is no normal equation, so we solve numerically

Methods: gradient descent, stochastic gradient descent (SGD), Newton's method.

Can add an L1 or L2 penalty to this problem convex while controlling overfitting.



OPTIMIZATION SETUP

OPTIMIZATION II: GRADIENT DESCENT

Gradient of the loss with respect to the weights:

$$\partial \mathcal{L}/\partial w = \Sigma_i \, (p_i - y_i) \, x_i$$

Update rule: step downhill, $w \leftarrow w - \eta \cdot \partial \mathcal{L}/\partial w$, where η is the learning rate.

A direct parallel to linear regression

- Linear regression: updates use residuals ($\hat{y} - y$)
- Logistic regression: updates use probabilistic errors ($p - y$)

Interpretation

- $p > y$: the gradient pushes the prediction down toward 0
- $p < y$: the gradient pulls the prediction up toward 1
- $p \approx y$: the signal is small — that observation is already explained

PART 3 — USING THE MODEL

From a fitted probability to a decision.

- How does a probability become a predicted class?
- Who decides the threshold, and on what grounds?
- Can you run the whole pipeline yourself?

CLASSIFICATION: THE DECISION RULE

Logistic regression outputs probabilities. Classification needs one more ingredient — a decision threshold τ:

$$\hat{y} = 1 \text{ if } P(y = 1 \mid x) \geq \tau, \quad \text{otherwise } \hat{y} = 0$$

The standard choice is τ = 0.5, which is equivalent to classifying on the sign of the score z.

Why τ = 0.5 is only a default

- It implicitly assumes a false positive and a false negative cost the same.
- Nothing in the fitting procedure chose it — the model never saw τ.
- Changing τ changes the predictions without refitting the model.

CHOOSING THE THRESHOLD

Set τ from the cost of each error type, not by habit.

- Medical screening: a missed case (false negative) is far costlier than a false alarm, so lower τ — catch more positives and accept more false positives
- Fraud review: each flag costs analyst time, so raise τ to protect precision
- Credit default: a lender who loses more on a bad loan than on a declined application should lower τ

Types of mistakes

False positive: predicted 1, actually 0. False negative: predicted 0, actually 1.

Sweeping τ from 0 to 1 traces the ROC and precision–recall curves — the subject of the next lecture.

CHECK YOUR UNDERSTANDING

A model fit to the Default data (balance in $1,000s) gives w = 2.0, b = −7.0. For a customer with a $2,000 balance (x = 2.0), work out the predicted probability and the class label.

Step 1 — compute the score

$$z = w^{T}x + b = 2.0(2.0) - 7.0 = -3.0$$

Step 2 — convert the score to a probability

$$p = \sigma(-3.0) = 1 \,/\, (1 + \exp(3.0)) \approx 0.047$$

Step 3 — apply the decision rule

$$p = 0.047 < \tau = 0.5 \;\Rightarrow\; \text{predict } \hat{y} = 0 \text{ (no default)}$$

Now try it yourself: what is the predicted class at a $4,000 balance, and what threshold would flip the $2,000 answer?

CONCEPTUAL SUMMARY

1. Start with the linear regression structure: $z = w^T x + b$
2. Interpret z as the log-odds (logit) of the outcome
3. Apply the sigmoid to turn the score into a probability
4. Model the outcome with a Bernoulli likelihood
5. Minimize the cross-entropy loss (convex, no closed form)
6. Solve with gradient-based optimization
7. Apply a threshold τ to turn probabilities into class decisions

You should now be able to

- Compute a predicted probability from w, b, and x by hand
- Interpret any coefficient as an odds ratio, and say why it is not a probability change
- Justify a threshold from the relative cost of the two error types